Living-ladder (Z') and cofinal R_0

Two compose remainders after r383

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Abstract

Round 386 attacks two named remainders of r383. (Z'): $|Z_{\text{loc}}| \le Z_0' < M$ on the *living* ladder (FRAME-A 89 + FRAME-B 8 + χ with $q_N < 1$). The six r383 χ-violators are exactly the terminal-dead sprouts $q_N > 1$. The death channel $|Z_{\text{loc}}| \ge M \iff q_N > 1$ holds on 181/181 as a biconditional census. The living supremum is 0.833870 at χ_4 kz46, which exceeds r383's $Z_0 = 4/5$ and forces $Z_0' = 21/25 = 0.84 < M$. Naive finite-head Chebyshev remains refuted (n_{edge} 167–4689; the edge triangle grows). Signed cancellation is real (median 0.055) but $\text{cancel}_{\max} \cdot \text{tri}_{\max} = 4.08 > M$, so a uniform-cancel times triangle does not close. (R): $B(\omega,\omega)/D \le 3/10$ on 181/181 (maximum 0.130, log-log slope -1.16). Cauchy–Schwarz on the $r298$ remainder is a theorem (181/181), tight at kz37. FAB $mq_{\max} \le 14.93 \log m$ is not family-uniform (FRAME-B FAB 18.07). CS+FAB therefore does not prove an $O(\text{polylog})$ bound on S_F/D. The named remainder is $B(\omega+\beta, \omega+\beta) \le KD$. $R_0 = 4$ stays a census. No RH claim.

Status. Lemma 1 is proved (triangle). Lemma 3 is proved (Fejér Gram). Lemma 2 is the r263 dictionary at the rho coordinate (gated at $w = 9$). Proposition 1 records (Z') reduced. Proposition 2 records cofinal R_0 reduced to a named Gram remainder. No RH claim.

1 Living (Z')

Write $M = \sqrt{5/7}$, $Z = Z_{\text{loc}} + t_{\text{bulk}}$ (r383, Lean `canonical_split`), and $q_N = \rho_{N-1}/(5/7)$. Under the r263 dictionary $Z^2 = (5/7)q_N$ one has $q_N < 1 \iff |Z| < M$.

Lemma 1 (Death triangle)**.** $|Z| \ge |Z_{\text{loc}}| - |t_{\text{bulk}}|$. *If* $|Z_{\text{loc}}| \ge M$ *and* $|t_{\text{bulk}}| \le |Z_{\text{loc}}| - M$, *then* $|Z| \ge M$.

Lemma 2 (Dictionary at rho)**.** *On FRAME-A* $w = 9$, $\rho_{N-1} = Z^2$ *and* $q_N = \rho_{N-1}/(5/7) = (7/5)Z^2$ *to relative* 10^{-15}.

Proposition 1 ((Z'), reduced)**.** *On the living ladder of* 175 *windows,* $|Z_{\text{loc}}| \le 21/25 < M$. *The measured supremum is* 0.833870 *at* χ_4 *kz*46. *The r383 constant* 4/5 *fails on that living* χ *window. The six dead windows are exactly* $\chi_3\{15, 19, 23, 33, 39\}$ *and* $\chi_4\{20\}$, *and on the* 181*-surface*

$$|Z_{\text{loc}}| \ge M \iff |Z| \ge M \iff q_N > 1.$$

This biconditional is a census, not a theorem: t_{bulk} *can reduce* $|Z|$ *relative to* $|Z_{\text{loc}}|$ *(living* χ_4*–*46*:* $0.834 \to 0.713$*) but does not pull any of the six dead rows below* M*. A PNT-free finite-head Chebyshev bound is refuted (*n_{edge} *grows; the unsigned edge triangle has log-log slope* $+0.25$ *and range* 2.86–14.51*). Signed cancellation is essential (median factor* 0.055*) and is not a uniform SATZ (*$\text{cancel}_{\max} \cdot \text{tri}_{\max} > M$*). A cofinal* m_2 *is not proved. Scramble seed* 1 *at* $w = 9$ *does not break* (Z').

Outcome: **REDUCED** to $Z_0' = 21/25$ on the living ladder, with the death channel a census iff and the source bound a named remainder (cancellation law, not a triangle).

2 Cofinal R_0

Lemma 3 (Fejér Cauchy–Schwarz). *$B(\Delta, \omega+\beta)^2 \le B(\Delta, \Delta)\, B(\omega+\beta, \omega+\beta)$. On 181/181 rows the inequality holds; at the R-star kz37 it is tight ($T_\Delta/D = 2.9116$ versus $\sqrt{B_{\Delta\Delta} B_\Sigma}/D \approx 2.921$).*

Proposition 2 (Cofinal R_0, reduced). *$S_F \le 4\,D$ remains a 181-census (maximum 2.916788 at FRAME-A kz37). The weaker window bound $B(\omega, \omega) \le (3/10)\,D$ holds on 181/181 (maximum 0.129976, median 0.010, log-log slope -1.16). A field-independent $C_{\mathrm{MAIN}} \le 1$ is refuted by the DC toy ($R = \mathrm{MAIN}/D = 3.6875$). CS+FAB does not yield $O((\log m)^B)$ on S_F/D: the A-law $mq_{\max} \le 14.93 \log m$ fails on FRAME-B (FAB 18.07), and FAB controls P_Δ not P_ω. The smallest named remainder is*

$$B(\omega + \beta, \omega + \beta) \le KD$$

for an absolute K (at the R-star one has $B_\Sigma/D = 2.975 < 4$ and $B_{\Delta\Delta}/D = 2.869 < 4$; CS then gives $|T_\Delta|/D \le 4$, and $S_F/D \le 3/10 + 4$). The log-log slope of R on the surface is -0.328 (census, not a theorem). Scramble seed 1 does not break $R_0 = 4$.

Outcome: **REDUCED**. Remaining: the Gram bound on $B(\omega+\beta, \omega+\beta)/D$ (or an equivalent weight-sum inequality on the window+border sum field).

3 Machine check

`verify_compose_premises2.py` and `compose_premises2_probe.py` gate the toys (DC C_{MAIN}-kill, period-2, CS, death triangle, Z_0' versus M), the $w = 9$ dictionary, scramble non-break, kz37 grams, living χ_4–46, dead χ_3–15, the 4/5 mutant, and on `--full` the 181-row census. Final line `COMPOSE PREMISES2 VERIFIED`. No RH claim.

References

[1] S. Hamann, *The compose premises* (R), (L), (Z), `rh/problem/compose_premises.pdf`, 2026.

[2] S. Hamann, *The compose lemma, reduced*, `rh/problem/compose_lemma.pdf`, 2026.

[3] `window_border_transfer_probe.py`.